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INSTITUTE OF SCIENCE AND TECHNOLOGY

Computer Organization and Assembly Language

Laboratory Manual

CSCL-3105

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- 1) Write a program to display a single character.
Write a program to input from with echo.
- 2) Write a program to input from without echo.
- 3) Write a program to display your full name on the screen.
- 5) Write a program which take four inputs from user without echo and display '*' (asterisk sign) in place of it (as in password).
- 6) Write a program which take upper-case single character as input from user and convert it into lower-case and display its result in the next line.
- 7) Write a program which take lower-case single character as input from user and convert it into upper-case and display its result in the next line.
- 8) Write a program to add two numbers.
- 9) Write a program to take two numbers as input from user, add them and display the result on screen in appropriate way (3+2=5).
- 10) Write a program to take two numbers as input from user, subtract them and display the result on screen in appropriate way (7 - 2=5).
- 11) Write a program to add three numbers using variables and stores in sum variable and display it.
- 12) Write a program to increment a number using increment (inc) instruction.
- 13) Write a program to decrement a number using decrement (dec) instruction.
- 14) Write a program to display "HELLO WORLD" using string variable.
- 15) Write a program to display full name, registration number, email address and Institute name.
- 16) Write a program to take times input from user using loop instruction.
- 17) Write a program to print numbers from '0' to '9' using loop.
- 18) Write a program to print numbers from '9' to '0' using loop.
- 19) Write a program to print upper case letter 'A' to 'Z' using loop.
- 20) Write a program to print lower case letter 'a' to 'z' using loop.
- 21) Write a program to print upper case letter 'Z' to 'A' using loop.
- 22) Write a program to print even numbers between (0-9) using loop.
- 23) Write a program to print" ##### "Using nested loop.
- 24) Write a program to print * using nested loop.
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- 25) Write a program to demonstrate the concept of an array. Use index operators to access the different values stored in an array.
- 26) Write a program to print an array using loop.
- 27) Write a program to take string input from user and store in an array.

- 28) Write a program to swap two numbers using stack.
- 29) Write a program to reverse the message "Hello world" using stack (push and pop instruction).
- 30) Write a program to take input and output using procedures.
- 31) Write a program to make procedures for newline and string display. Display your full name, registration number, email and institute name by using nested procedures.
- 32) Write an assembly code to take multiple character as input from user with prompt and program should tell how many vowels and consonants were there in the input. Use enter key to display the both number of counts with prompt. After displaying the result, ask him again for a choice that he wants to run program again or terminate the program. Use "Y" to run again and Use "Esc key" to terminate the program.
- 33) Write a program which take upper-case single character as input from user and convert it into lower-case and display its result in the next line using logic instruction.
- 34) Write a program which take lower-case single character as input from user and convert it into upper-case and display its result in the next line using logic instruction.
- 35) Write a program which takes string input from user and converts that string into lower case, upper case and toggle case using logic instruction only.
- 36) Write a password program which take input from user and display '*' on screen. Then compare the input character with the store character and display "Correct Password" if password correctly enter else print "Incorrect password" on screen.
- 37) Write an assembly program to multiply two numbers.
- 38) Write a program to take two number as input from user then multiply them and display the result after multiplication.
- 39) Write a programme to divide two numbers.
- 40) Write a program to take two number as input from user then divide them and display the result after division.
- 41) Write a program which ask user to enter a digit (from 1 to 9 only) and then print the table of that number.

The program is look like this

Please enter a digit = 3

(User press 3 and then press Enter key)

3 x 01 = 03

3 x 02 = 06

3 x 03 = 09

3 x 04 = 12

3 x 05 = 15

3 x 06 = 18

3 x 07 = 21

3 x 08 = 24

3 x 09 = 27

3 x 10 = 30

43) Write a program that checks whether the entered number is even or odd using div.

44) Write a program that take input as string and counts total number of characters entered without spaces (spacebar) and display the result when user press 'ENTER' key it must show the result not including the 'ENTER' key. Use macro for string display and new line.

45) Write a program to take input from user, check and informing user what he/she entered. Use macro for string display and new line.

- a) upper case letter
- b) lower case letter
- c) number
- d) symbol